

BZFlag Renderer Modernization - Final Summary

Project: drag BZFlag 2.4's legacy fixed-function renderer into the 21st century. Constraints: no new libraries, ARM64 (Adreno X1-85, GL 4.6 Mesa 26), compatibility profile, incremental in-repo upgrade. Baseline: 240 glBegin calls in 51 files, 46 display lists, zero shaders, zero VBOs. All work shipped as installable debs (latest: bzflag 2.4.31-9).

Tier 1 - Low-Hanging Visual Upgrades (deb 2.4.31-6)

- MSAA anti-aliasing: SDL2 visual multisample support, default 4x (was off)
- Anisotropic texture filtering: default 8x (was 1), clamped to GPU max, also applied to fonts
- Max texture size default raised 1024 to 4096
- Lighting: specular scaled to 0.5 (kills plastic blow-out); tank material specular 0.4 / shininess 40 (painted-metal highlight)
- Ambient light raised with blue tint at night; softer day ambient
- Fog: EXP2 mode, sky-tinted fog color replacing flat dark grey; slightly denser default fog
- MSAA framebuffer RGBA fix (control panel blending)
- Textures rebuilt: roof.png 64px noise to 512px seamless cement tile; moon.png 128 to 512 Lanczos upscale

Tier 2 Stage A - Tank Geometry on VBOs (deb 2.4.31-2/3)

- All 14 tank model files converted from display lists to VBOs (~450 small VBOs replacing ~450 display lists)
- Interleaved pos+normal+uv batches (8 floats/vertex) drawn via glDrawArrays
- Fan primitives re-tessellated to triangles (glDrawArrays has no fan mode)
- Zero visual change - pure draw-path swap, fixed-function state untouched
- Fixed capture bug where other tanks were invisible (vertices dropped)

Tier 2 Stage B - GLSL Per-Pixel Tank Lighting (deb 2.4.31-5)

- New TankLightingShader: GLSL 1.20, per-pixel Blinn-Phong replacing per-vertex Gouraud
- Reads built-in gl_LightSource / gl_FrontMaterial / gl_Fog - no CPU state mirroring needed
- Dynamic light mask + fog mode passed as uniforms; automatic fallback to fixed function on compile/link failure

Tier 2 Stage C - Sun Shadow Mapping (deb 2.4.31-6)

- New ShadowMapFBO: depth-only FBO (GL_DEPTH_COMPONENT32), size from new BZDB var shadowMapSize (default 2048, 64-4096)
- New ShadowMapper: per-frame sun-view depth pass over the whole map, ortho projection
- 4-tap PCF shadow softening, normal-offset bias; out-of-bounds = lit

- Shadow factor multiplies sun diffuse+specular only (dynamic lights unaffected)
- Graceful fallback to old stippled ground shadows when gated off

Tier 3 Stage A - Radar De-Immediate-Mode (deb 2.4.31-7/8/9)

- New GLBatch utility: drop-in glBegin/glEnd replacement, flushes via glDrawArrays, fans expanded, line-loops closed, GL current-color semantics respected (glGetFloatv at begin)
- Radar cluster at zero glBegin: RadarRenderer (walls, boxes, pyramids, mesh fans, outlines, bases, teleporters, noise, view wedge, north marker, tick, tank/flag markers) plus 3 shot strategies (SegmentedShot, GuidedMissile, ShockWave)
- Fixed white-blocks bug: root cause was a stale binary in the -8 deb (GLBatch fix never linked); verified live in-game on a local bzfs server - radar obstacles now render teal/cyan as intended (-9)
- Hardening: RadarRenderer re-issues the tank color before height-box batches (colorScratch)

Packaging & CI

- Standalone deb package: binaries, desktop entry, hicolor 256+48 icons, postinst/postrm hooks - verified on every build
- Package renamed bzflag21 back to bzflag (Conflicts/Replaces handles upgrades both ways)
- GitHub Actions workflow builds the deb ARM64 on every push and attaches it to releases on tag push
- Repo published: github.com/PC-Support-Services/bzflag-arm64-sp, branch 2.4

Verification

- make check clean at every stage
- Every deb verified: dpkg status, installed binary cmp-matches build, icons + desktop present
- Tier 2 stages user-tested visually (VBO tanks, lighting, shadows confirmed good)
- Tier 3 radar fix verified live: local bzfs server, scripted client join, screenshots before (white blocks) and after (teal shapes), GLBATCH_DEBUG color trace

Remaining (Tier 3 continuation)

- Stage B: HUD cluster (~30 glBegin sites)
- Stage C: world/effects cluster (~69 sites)
- Stage D: debug/rare paths (~29 sites)

Deliverables

- Installed: bzflag 2.4.31-9 (arm64) - /usr/local/bin/{bzflag,bzfs,bzadmin}
- Package: /home/richard/Documents/projects/bzflag/bzflag_2.4.31-9_arm64.deb
- Repo: github.com/PC-Support-Services/bzflag-arm64-sp (branch 2.4)
- Change log: history.md in the project folder